

Lecture 19: Normal Matrices

Hermitian Geometry, Spectral Projections, and Orthogonal
Eigenspaces

The Problem

Today's Question

Previous lectures gave spectral decomposition:

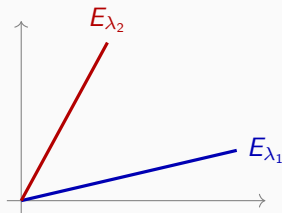
$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k.$$

This splits the whole space into eigenspace pieces:

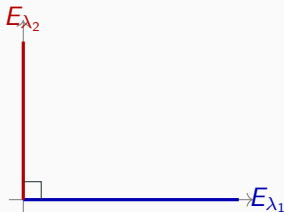
$$\mathbb{C}^n = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}.$$

When is this direct sum actually orthogonal?

Oblique vs. Orthogonal Eigenspaces



oblique decomposition



orthogonal decomposition

Normal matrices are exactly the matrices whose spectral pieces are orthogonal.

The Main Theorem

Theorem (Normal Matrix Theorem)

For a complex matrix A ,

A is normal $\iff$ A is diagonalizable
and distinct eigenspaces are Hermitian-orthogonal.

In spectral projection language:

$$A \text{ normal} \iff P_i^H = P_i \text{ for all } i.$$

Diagonalizable is not the endpoint. Orthogonal eigenspaces are the endpoint.

Hermitian Geometry

Why Complex Geometry Needs Repair

Over real numbers, length comes from

$$\|v\|^2 = v^T v.$$

However, problem arises when working over complex numbers.

Take

$$v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Then

$$v^T v = 1 + i^2 = 0.$$

A nonzero vector should not have length zero.

The Repair: Conjugate First

The complex mirror of a vector is

$$v^H = \overline{v}^T.$$

For

$$v = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad v^H = \begin{pmatrix} 1 & -i \end{pmatrix}.$$

Now

$$v^H v = 1 + (-i)i = 2.$$

Complex length is Hermitian length.

Hermitian Inner Product

Definition (Hermitian Inner Product)

For $v, w \in \mathbb{C}^n$, define

$$\langle v, w \rangle_H = v^H w.$$

Then

$$\langle v, v \rangle_H = v^H v = |v_1|^2 + |v_2|^2 + \cdots + |v_n|^2 \geq 0.$$

And

$$v^H v = 0 \iff v = 0.$$

Hermitian Orthogonality

Definition (Hermitian Orthogonal Vectors)

Two vectors $v, w \in \mathbb{C}^n$ are **Hermitian-orthogonal** if

$$v^H w = 0.$$

We write

$$v \perp_H w.$$

Definition (Hermitian-Orthogonal Subspaces)

Subspaces $V, W \subseteq \mathbb{C}^n$ satisfy $V \perp_H W$ if

$$v^H w = 0 \quad \text{for every } v \in V, \quad w \in W.$$

Hermitian Direct Sum

When eigenspaces are Hermitian-orthogonal, we write

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

This means every vector has a unique decomposition

$$v = v_1 + v_2 + \cdots + v_k, \quad v_i \in E_{\lambda_i},$$

and different pieces are perpendicular:

$$v_i^H v_j = 0 \quad (i \neq j).$$

This is the geometric meaning of normality.

Hermitian Conjugate and Normality

Hermitian Conjugate of a Matrix

Definition (Hermitian Conjugate)

For a complex matrix A , define

$$A^H = \overline{A}^T.$$

Example:

$$A = \begin{pmatrix} 1 & i \\ 2-i & 3 \end{pmatrix}, \quad A^H = \begin{pmatrix} 1 & 2+i \\ -i & 3 \end{pmatrix}.$$

If A is real, then $A^H = A^T$.

Product Rule

Hermitian conjugate reverses multiplication order:

$$(AB)^H = B^H A^H.$$

So H behaves like transpose, but with complex conjugation added.

$$\boxed{A} \quad \boxed{B} \xrightarrow{H} \boxed{B^H} \quad \boxed{A^H}$$

Definition (Normal Matrix)

A complex square matrix A is **normal** if

$$AA^H = A^H A.$$

It means exactly this equation: A commutes with A^H .

Special Normal Matrices

Name	Equation	Example
Hermitian	$A^H = A$	$\begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$
Skew-Hermitian	$A^H = -A$	$\begin{pmatrix} i & 1 \\ -1 & 2i \end{pmatrix}$
Unitary	$A^H A = I$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

These are not separate topics. They are special normal matrices.

Non-Example

Let

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$JJ^H = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad J^H J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

So

$$JJ^H \neq J^H J.$$

J is not normal. Its eigen-directions are not orthogonal geometry.

Proof Plan

Recall: Radical Criterion

Write the characteristic polynomial honestly as

$$\det(tI - A) = \prod_{i=1}^k (t - \lambda_i)^{m_i}.$$

The radical polynomial removes repeated powers:

$$r_A(t) = \prod_{i=1}^k (t - \lambda_i).$$

From previous lectures:

Theorem (Radical Criterion for Diagonalizability)

For a complex matrix A ,

$$A \text{ diagonalizable} \iff r_A(A) = 0.$$

This is the criterion we are trying to force for normal matrices.

Goal and Subgoals

Goal:

$$A \text{ normal} \implies A \text{ diagonalizable.}$$

By the recalled radical criterion,

$$A \text{ diagonalizable} \iff r_A(A) = 0.$$

So the proof goal becomes:

$$r_A(A) = 0.$$

Subgoal	Input assumption	Output
Theorem A	any complex matrix A	$r_A(A)$ is nilpotent
Theorem B	A normal, $p(t)$ any polynomial	$p(A)$ is normal
Lemma C	N normal and nilpotent	$N = 0$

Important boundary: Theorem A is true even if A is not normal.

How the Subgoals Prove the Goal

Now specialize to a normal matrix A :

$$\underbrace{r_A(A) \text{ is nilpotent}}_{\text{Theorem A: true for every } A} \quad \text{and} \quad \underbrace{r_A(A) \text{ is normal}}_{\text{Theorem B: uses normality of } A} .$$

Then

$$\underbrace{r_A(A) = 0}_{\text{Lemma C}} .$$

From the previous diagonalization criterion:

$$r_A(A) = 0 \implies A \text{ is diagonalizable.}$$

After diagonalizability, spectral projections will prove orthogonality.

Theorem A: $r_A(A)$ Is Nilpotent

Theorem (Radical Polynomial Produces Nilpotent Matrix)

For every complex matrix A , the matrix $r_A(A)$ is nilpotent.

Let

$$M = \max\{m_1, \dots, m_k\}.$$

Then

$$r_A(t)^M = \prod_{i=1}^k (t - \lambda_i)^M = \det(tI - A)q(t).$$

Plug in A :

$$r_A(A)^M = 0.$$

Theorem B: Polynomials Preserve Normality

Theorem (Polynomial of a Normal Matrix)

If A is normal, then every polynomial $p(A)$ is normal.

Normality says

$$AA^H = A^H A.$$

So every power of A commutes with every power of A^H :

$$A^m (A^H)^n = (A^H)^n A^m.$$

Therefore $p(A)$ commutes with $p(A)^H$.

Positivity Lemma for Matrices

Lemma (Matrix Positivity)

If $MM^H = 0$, then $M = 0$.

List the rows of M as

$$M = \begin{pmatrix} - & r_1^H & - \\ - & r_2^H & - \\ & \vdots & \\ - & r_m^H & - \end{pmatrix}.$$

Then MM^H is the inner-product table of the rows.

Positivity Lemma: Inner-Product Table

$$\underbrace{\begin{pmatrix} - & r_1^H & - \\ - & r_2^H & - \\ & \vdots & \\ - & r_m^H & - \end{pmatrix}}_M \underbrace{\begin{pmatrix} | & | & & | \\ r_1 & r_2 & \cdots & r_m \\ | & | & & | \end{pmatrix}}_{M^H} = \underbrace{\begin{pmatrix} r_1^H r_1 & r_1^H r_2 & \cdots & r_1^H r_m \\ r_2^H r_1 & r_2^H r_2 & \cdots & r_2^H r_m \\ \vdots & \vdots & \ddots & \vdots \\ r_m^H r_1 & r_m^H r_2 & \cdots & r_m^H r_m \end{pmatrix}}_{MM^H}.$$

If $MM^H = 0$, then in particular every diagonal entry is zero:

$$r_i^H r_i = 0 \implies r_i = 0.$$

So every row is zero, hence $M = 0$.

Lemma C: Normal Nilpotent Collapse

Lemma (Normal Nilpotent Matrices Vanish)

If N is normal and nilpotent, then

$$N = 0.$$

This is where the matrix positivity lemma enters:

$$MM^H = 0 \implies M = 0.$$

We prove this by descending from a high zero power down to $N = 0$.

Lemma C: Descent Step

Assume

$$N^{2^r} = 0.$$

Since N is normal, N commutes with N^H .

Then write the power as repeated factors:

$$0 = \underbrace{NN \cdots N}_{2^r \text{ many}} \underbrace{N^H N^H \cdots N^H}_{2^r \text{ many}}.$$

Because N commutes with N^H , regroup this as:

$$0 = \underbrace{\left(\underbrace{NN \cdots N}_{2^{r-1} \text{ many}} \underbrace{N^H N^H \cdots N^H}_{2^{r-1} \text{ many}} \right)}_M \underbrace{\left(\underbrace{NN \cdots N}_{2^{r-1} \text{ many}} \underbrace{N^H N^H \cdots N^H}_{2^{r-1} \text{ many}} \right)^H}_{M^H}.$$

By the matrix positivity lemma,

$$N^{2^{r-1}} (N^H)^{2^{r-1}} = 0.$$

Lemma C: Finish

Repeat the descent:

$$N^{2^r} = 0 \Rightarrow N^{2^{r-1}}(N^H)^{2^{r-1}} = 0 \Rightarrow \cdots \Rightarrow NN^H = 0.$$

By positivity again:

$$NN^H = 0 \quad \Longrightarrow \quad N = 0.$$

Normal plus nilpotent leaves no room except zero.

Normal Implies Diagonalizable

For normal A :

- Theorem A: $r_A(A)$ is nilpotent.
- Theorem B: $r_A(A)$ is normal.
- Lemma C: normal nilpotent matrices vanish.

Therefore

$$r_A(A) = 0.$$

By the radical criterion:

A is diagonalizable.

But this is only the checkpoint, not the endpoint.

Spectral Projections

Spectral Projections Return

For a diagonalizable matrix with distinct eigenvalues λ_i , previous lectures gave

$$P_i = f_i(A),$$

where f_i is the Lagrange interpolation polynomial satisfying

$$f_i(\lambda_j) = \delta_{ij}.$$

Then

$$I = P_1 + \cdots + P_k, \quad P_i P_j = 0 \ (i \neq j), \quad A = \sum_i \lambda_i P_i.$$

What Does P_i Do?

The projection P_i extracts the λ_i -eigenspace component:

$$\text{Im}(P_i) = E_{\lambda_i}.$$

Every vector decomposes as

$$v = P_1 v + P_2 v + \cdots + P_k v.$$

Question:

Are these projections oblique or Hermitian-orthogonal?

For normal matrices, they are Hermitian-orthogonal projections.

Normality Passes to Spectral Projections

If A is normal, then by Theorem B every polynomial in A is normal.

But

$$P_i = f_i(A).$$

Therefore

$$P_i \text{ is normal.}$$

Also, by construction,

$$P_i^2 = P_i.$$

So each spectral projection is a normal projection.

The Central Lemma

Lemma (Normal Projection Is Hermitian)

If

$$P^2 = P \quad \text{and} \quad PP^H = P^H P,$$

then

$$P = P^H.$$

Normality kills obliqueness for projections.

Proof: Algebra Setup

We prove $P = P^H$ by proving

$$P - PP^H = 0.$$

Let

$$M = P - PP^H.$$

By the matrix positivity lemma, it is enough to show

$$MM^H = 0.$$

Everything now is algebra: expand and simplify.

Proof: Expand MM^H

Since $(PP^H)^H = PP^H$,

$$M^H = P^H - PP^H.$$

So

$$\begin{aligned} MM^H &= (P - PP^H)(P^H - PP^H) \\ &= PP^H - P(PP^H) - (PP^H)P^H + (PP^H)(PP^H). \end{aligned}$$

Now use $P^2 = P$, $(P^H)^2 = P^H$, and $PP^H = P^H P$.

Proof: Simplify

Term by term:

$$P(PP^H) = P^2P^H = PP^H.$$

$$(PP^H)P^H = P(P^H)^2 = PP^H.$$

For the last term:

$$(PP^H)(PP^H) = P(P^HP)P^H = P(PP^H)P^H = P^2(P^H)^2 = PP^H.$$

Proof: Finish

Substitute these into the expansion:

$$MM^H = PP^H - PP^H - PP^H + PP^H = 0.$$

By the matrix positivity lemma:

$$MM^H = 0 \implies M = 0.$$

So

$$P - PP^H = 0, \quad \text{hence} \quad P = PP^H.$$

Taking H gives $P^H = PP^H$, so $P = P^H$.

Projection Payoff

Every spectral projection of a normal matrix satisfies

$$P_i^2 = P_i, \quad P_i P_i^H = P_i^H P_i.$$

By the central lemma:

$$P_i = P_i^H.$$

This is where orthogonal eigenspaces come from.

For normal A :

- A is diagonalizable.
- $P_i = f_i(A)$ are spectral projections.
- P_i are normal because polynomials preserve normality.
- $P_i^2 = P_i$ because they are projections.
- Therefore $P_i = P_i^H$.

So every spectral projection is Hermitian symmetric.

Orthogonal Eigenspaces

If $P_i = P_i^H$, then P_i is an orthogonal projection.

For $i \neq j$,

$$P_i P_j = 0.$$

Take $v \in E_{\lambda_i} = \text{Im}(P_i)$ and $w \in E_{\lambda_j} = \text{Im}(P_j)$.

Then

$$v^H w = (P_i v)^H (P_j w) = v^H P_i^H P_j w = v^H P_i P_j w = 0.$$

Distinct eigenspaces are Hermitian-orthogonal.

Converse Direction

Assume A is diagonalizable and all spectral projections are Hermitian:

$$A = \sum_i \lambda_i P_i, \quad P_i^H = P_i, \quad P_i P_j = 0 \ (i \neq j).$$

Then

$$A^H = \sum_i \bar{\lambda}_i P_i.$$

So

$$AA^H = \sum_i |\lambda_i|^2 P_i = A^H A.$$

Therefore A is normal.

Main Theorem, Proven

Theorem (Normal Matrix Theorem)

For a complex matrix A ,

A normal $\iff$ A is diagonalizable
and distinct eigenspaces are Hermitian-orthogonal.

Equivalently:

$$A \text{ normal} \iff P_i = P_i^H \text{ for all } i.$$

Unitary Diagonalization

Collect Orthonormal Eigenvectors

Because eigenspaces are Hermitian-orthogonal, we can choose an orthonormal basis inside each eigenspace and stack all of them:

$$\Omega = \begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & \cdots & | \end{pmatrix}.$$

Then

$$\Omega^H \Omega = I.$$

So Ω is unitary.

Unitary Diagonalization

With the orthonormal eigenbasis matrix Ω ,

$$A\Omega = \Omega\Lambda.$$

Multiply by Ω^H :

$$\Omega^H A \Omega = \Lambda.$$

Equivalently,

$$A = \Omega \Lambda \Omega^H.$$

Unitary diagonalization is a consequence of orthogonal eigenspaces.

Complete Example

Example: Start from a Hermitian Matrix

Let

$$\underbrace{\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}}_A.$$

This matrix is Hermitian symmetric:

$$A^H = A.$$

Therefore:

$$A \text{ Hermitian} \implies A \text{ normal} \implies A \text{ diagonalizable.}$$

Example: Radical Criterion

Compute the characteristic polynomial:

$$\det(tI - A) = t(t + 3)^2.$$

So the radical polynomial is

$$r_A(t) = t(t + 3).$$

Since A is normal, it is diagonalizable. Therefore the radical criterion gives:

$$r_A(A) = A(A + 3I) = 0.$$

Now we can use the spectral-projection recipe from previous lectures.

Example: Polynomial Remainder Formula

For any polynomial $g(t)$, reduce g modulo

$$r_A(t) = t(t + 3).$$

The two eigenvalues are 0 and -3 .

By Lagrange interpolation:

$$g(A) = g(0) \underbrace{\frac{A + 3I}{3}}_{P_0} + g(-3) \underbrace{\frac{-A}{3}}_{P_{-3}}.$$

So the spectral projections are:

$$\boxed{P_0 = \frac{A + 3I}{3}}, \quad \boxed{P_{-3} = \frac{-A}{3}}.$$

Example: Honest Projection Matrices

$$P_0 = \frac{1}{3} \left(\underbrace{\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}}_A + 3I \right) = \frac{1}{3} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{P_0}.$$

$$P_{-3} = \underbrace{I - P_0}_{P_{-3}} = \frac{1}{3} \underbrace{\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}}_{P_{-3}}.$$

Example: Projection Decomposition

The spectral decomposition is:

$$\underbrace{\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}}_A = 0 \cdot \underbrace{\frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{P_0} - 3 \cdot \underbrace{\frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}}_{P_{-3}}.$$

The projections are Hermitian, so their column spaces are orthogonal.

Example: Diagonal Cross-Fill P_0

The projection P_0 is rank one:

$$P_0 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}}_{u_1 u_1^H}.$$

So one unit eigenvector for eigenvalue 0 is

$$u_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Example: Diagonal Cross-Fill P_{-3}

Now cross-fill the orthogonal projection P_{-3} .

$$\underbrace{\frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}}_{P_{-3}} = \underbrace{\begin{pmatrix} \sqrt{\frac{2}{3}} \\ -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} \sqrt{\frac{2}{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{pmatrix}}_{u_2 u_2^H} + \underbrace{\begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{u_3 u_3^H}.$$

Example: One Equality Chain

The diagonalization is not a separate trick. It is the same spectral decomposition, after cross-filling the projections:

$$\begin{aligned}
 \underbrace{\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}}_A &= \underbrace{0 \underbrace{u_1 u_1^H}_{P_0}}_{\text{rank-one part of } P_{-3}} \underbrace{(-3) \underbrace{u_2 u_2^H}_{\text{rank-one part of } P_{-3}}} \underbrace{(-3) \underbrace{u_3 u_3^H}_{\text{rank-one part of } P_{-3}}} \\
 &= \underbrace{\begin{pmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{pmatrix}}_{\Omega} \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}}_{\Lambda} \underbrace{\begin{pmatrix} - & u_1^H & - \\ - & u_2^H & - \\ - & u_3^H & - \end{pmatrix}}_{\Omega^H} \\
 &= \boxed{\Omega \Lambda \Omega^H}.
 \end{aligned}$$

$$\Omega = \begin{pmatrix} \frac{1}{\sqrt{3}} & \sqrt{\frac{2}{3}} & 0 \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad \Omega^H \Omega = I.$$

Special Normal Matrices

Eigenvalue Geometry

For a normal matrix,

$$A = \sum_i \lambda_i P_i, \quad P_i = P_i^H.$$

Then

$$A^H = \sum_i \overline{\lambda_i} P_i.$$

So any extra equation involving A^H becomes an equation about eigenvalues.

Special normal matrices are eigenvalue-location restrictions.

Hermitian Matrices

Suppose

$$A^H = A.$$

Compare spectral decompositions:

$$\sum_i \bar{\lambda}_i P_i = \sum_i \lambda_i P_i.$$

Therefore

$$\bar{\lambda}_i = \lambda_i.$$

So

$$\boxed{\lambda_i \in \mathbb{R}.$$

Hermitian means: eigenvalues lie on the real axis.

Skew-Hermitian Matrices

Suppose

$$A^H = -A.$$

Compare spectral decompositions:

$$\sum_i \bar{\lambda}_i P_i = - \sum_i \lambda_i P_i.$$

Therefore

$$\bar{\lambda}_i = -\lambda_i.$$

So

$$\lambda_i \in i\mathbb{R}.$$

Skew-Hermitian means: eigenvalues lie on the imaginary axis.

Unitary Matrices

Suppose

$$A^H A = I.$$

Using

$$A = \sum_i \lambda_i P_i, \quad A^H = \sum_i \overline{\lambda_i} P_i,$$

we get

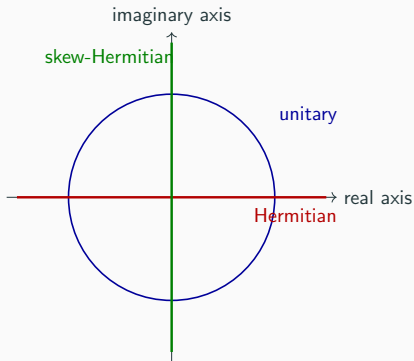
$$A^H A = \sum_i |\lambda_i|^2 P_i.$$

Thus

$$|\lambda_i|^2 = 1.$$

Unitary means: eigenvalues lie on the unit circle.

Complex Plane Summary



Matrix type	Eigenvalue location
Hermitian	real axis
Skew-Hermitian	imaginary axis
Unitary	unit circle

Final Summary

The core theorem is:

$$\boxed{AA^H = A^H A} \iff \boxed{P_i = P_i^H \text{ for all } i.}$$

$$\iff \boxed{E_{\lambda_i} \perp_H E_{\lambda_j} \quad (i \neq j).}$$

Special normal matrices add eigenvalue restrictions:

$$A^H = A \iff \lambda_i \in \mathbb{R},$$

$$A^H = -A \iff \lambda_i \in i\mathbb{R},$$

$$A^H A = I \iff |\lambda_i| = 1.$$

Normal matrices turn spectral decomposition into orthogonal geometry.